

Unit IV

Introduction to Automation and Control

Sensors and Transducers: Introduction to Sensors, transducers and their applications, basics of Feedback and Control: Open-loop vs Closed-loop Systems. **Overview of Mechanical Actuation System:** Overview of kinematic chains, cam mechanisms, ratchet mechanisms, gears and their types, belts, and bearings. **Hydraulic and Pneumatic Actuation Systems:** Overview of fluid power systems, Pressure and direction control valves, rotary actuators, basic hydraulic and pneumatic sequencing problems.

1. Sensors and Transducers

Sensors: A **sensor** is a device that detects changes in a physical quantity (like temperature, pressure, displacement, force, light, etc.) and converts it into a measurable signal, often electrical. Sensors are the “eyes and ears” of an engineering system, enabling machines to sense the environment. **Examples:** Thermometer (senses temperature), Microphone (senses sound pressure), Photodiode (senses light intensity).

Transducers: A **transducer** is a device that converts one form of energy into another. All sensors are transducers, but not all transducers are sensors. **Examples:** Thermocouple: Heat energy → Voltage, Loudspeaker: Electrical energy → Sound, Strain Gauge: Mechanical strain → Change in resistance

Basic Difference b/w Sensor and Transducer:

- **Sensor:** detects a signal/quantity.
- **Transducer:** converts it into another form, often electrical for further processing.

1.3 Types of Transducers

Active Transducers

- Generate their own output signal in response to input stimulus.
- Do not require external power.
- Example: Piezoelectric crystal (pressure → voltage).

Passive Transducers

- Require external power supply to operate.
- Input stimulus changes some property (resistance, capacitance, inductance).
- Example: Thermistor (temperature → resistance change).

1.4 Characteristics of Sensors/Transducers

When selecting or designing a sensor/transducer for a measurement system, it is important to understand its performance characteristics. These characteristics determine how well the device can detect, measure, and respond to the physical quantity being monitored.

Accuracy: Accuracy refers to the degree of closeness of a measured value to the true or actual value of the quantity. A highly accurate sensor provides values very close to the actual (true) value. Errors such as calibration errors, environmental effects, and noise can affect accuracy. **Example:** If the true temperature is **100 °C**, and a temperature sensor reads **99.8 °C**, the accuracy is high. But if another sensor reads **96 °C**, it is less accurate.

Precision: Precision is the ability of a sensor to consistently reproduce the same measurement under unchanged conditions, i.e., repeatability. A sensor may not be accurate but can still be precise if it gives the same reading every time for the same input. **Example:** Suppose the true pressure is **50 bar**. Sensor A gives readings: 49.9, 50.0, 50.1 (high precision, also accurate). Sensor B gives readings: 45.0, 45.1, 45.0 (high precision, but not accurate).

Sensitivity: Sensitivity is the ratio of change in output signal to the corresponding change in input signal. It represents how small a change in input the sensor can detect. Higher sensitivity means the sensor can detect very small changes. **Example:** A thermocouple generates **40 $\mu\text{V}/^\circ\text{C}$** . If the input changes by **1 °C**, the output changes by **40 μV** . A more sensitive thermocouple (**60 $\mu\text{V}/^\circ\text{C}$**) can detect even smaller temperature changes more effectively.

Linearity: Linearity is the degree to which the sensor's output is directly proportional to the input over its specified range. Ideally, the output vs. input graph should be a straight line. Deviations from linearity introduce measurement errors. **Example:** A strain gauge converts force to voltage. If doubling the applied force doubles the output voltage, the sensor is linear. If not, calibration and correction may be required.

Range: The range is the span between the minimum and maximum values of the input variable that a sensor can measure accurately. A sensor should be selected according to the expected range of the measurement application. **Example:** A pressure sensor with a range of **0–10 bar** cannot measure 50 bar. A thermocouple designed for **0–1000 °C** is unsuitable for cryogenic applications (e.g., **-100 °C**).

Response Time: Response time is the time taken by a sensor to respond to a change in the input and reach a certain percentage (commonly 95%) of the final steady-state output. A faster response time is crucial in dynamic systems where inputs change rapidly. **Example:** An infrared temperature sensor measuring furnace temperature may have a response time of **10 ms** (fast). A mercury-in-glass thermometer may take **30–60 seconds** to stabilize (slow).

Characteristic	Meaning	Example
Accuracy	Closeness to true value	True temp = 100 °C, sensor = 99.8 °C
Precision	Repeatability	Sensor always shows 45.0 °C for same true 50 °C
Sensitivity	Output change per unit input	Thermocouple: 40 $\mu\text{V}/^\circ\text{C}$

	change	
Linearity	Proportional input-output relation	Force vs. voltage in strain gauge
Range	Min–max measurable limits	Pressure sensor: 0–10 bar
Response Time	Speed of response to change	IR thermometer: 10 ms

Applications of Sensors and Transducers:

- **Industrial Automation:** Pressure, flow, temperature sensors in process industries.
- **Medical Field:** ECG, EEG, glucose sensors.
- **Automobiles:** Oxygen sensor, speed sensor, parking sensors.
- **Aerospace:** Gyroscopes, accelerometers.
- **Consumer Electronics:** Touch screens, microphones, cameras.

2. Feedback and Control Systems

Control System: A control system is an arrangement of physical components designed to regulate, command, direct, or control the behavior of other systems. Its purpose is to achieve the desired output in the presence of disturbances and uncertainties.

For example, in an air-conditioner, the thermostat acts as a control system to maintain the room temperature at a set value regardless of changes in outside temperature.

“A control system is a set of devices or mechanisms that manage, regulate, and control the behavior of a process or plant so that the output follows the desired input (reference signal)”

Elements of a Control System

Input / Reference Signal: The desired value or target that the system should achieve. It represents the setpoint or command given to the system. **Example:** In cruise control of a car, the reference input is the desired speed (e.g., 80 km/h). In an air-conditioner, the set temperature (e.g., 24 °C) is the reference input.

Process / System / Plant: The actual physical system or process being controlled. It transforms the input (with the effect of control action) into an output. **Example:** In a car’s cruise control, the **car engine + vehicle dynamics** act as the plant. In temperature control, the **room + heater/cooler** is the plant.

Output: The actual response or behavior of the system after applying the input and control actions. Ideally, the output should match the input/reference signal. **Example:** In a washing machine, the output could be the actual wash cycle time or water temperature. In a robotic arm, the output is the actual position of the arm.

Controller: The controller is the decision-making unit of the system. It compares the reference input with the output (via feedback) and decides the necessary corrective action. The controller generates a **control signal** to drive the plant toward the desired output. Controllers can be electronic circuits, mechanical devices, or software algorithms. **Example:** In cruise control, the controller adjusts the throttle to maintain constant speed. In a room heater, the thermostat acts as a simple ON/OFF controller. Modern control uses **PID controllers** (Proportional, Integral, Derivative) for better accuracy.

Feedback Mechanism: Feedback is the process of measuring the output and feeding it back to the input for comparison. This helps in reducing errors and disturbances. **Example:** In air-conditioning, a temperature sensor measures the actual room temperature and feeds it back to the controller (thermostat). In an automatic voltage regulator (AVR), the actual generator voltage is compared with the reference voltage to maintain stable output.

Negative feedback: Reduces error and stabilizes the system. (Most control systems use this.)

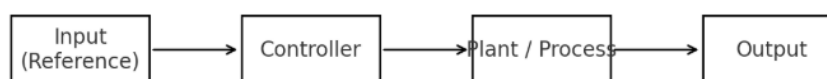
Positive feedback: Enhances or reinforces changes; rarely used in control due to instability.

Illustrative Example: Room Temperature Control

- **Input/Reference:** Set temperature = 24 °C
- **Process/Plant:** Room + air conditioner
- **Output:** Actual room temperature
- **Controller:** Thermostat controlling compressor ON/OFF
- **Feedback:** Temperature sensor reporting current temperature to thermostat

Open-loop System

An **open-loop system** is a control system in which the control action is independent of the actual output. The system does not measure the output, so it cannot compare it with the desired input or make corrections.



In open-loop system no feedback path from output to input is required. The system works on predefined settings only.

Examples: **Electric Iron:** Heats according to the preset thermostat value, but does not sense the actual iron temperature continuously. **Washing Machine Timer:** Runs the wash cycle for a fixed duration irrespective of the actual cleanliness of clothes. **Toaster:** Heats for a fixed time regardless of how toasted the bread actually becomes.

Advantages

- **Simple and inexpensive:** No sensors or feedback needed.
- **Easy to design and implement:** Fewer components and less complexity.
- **Faster response:** Since no feedback loop is present.

Disadvantages

- **Less accurate:** Output may deviate from the desired value due to disturbances.
- **No correction possible:** Cannot detect or correct errors automatically.
- **Performance depends on calibration:** Works well only if disturbances are minimal and system is predictable.

Closed-loop System (Feedback System)

A **closed-loop system** (or feedback system) is a control system in which the output is continuously monitored using sensors, and this feedback is compared with the desired input. The controller then adjusts the control action to minimize the error.

In closed-loop system, **Error = Input – Output (feedback)**. The controller modifies its action based on this error.

Examples:

1. Air Conditioner Thermostat:

Input: Desired room temperature (set by user).

Output: Actual room temperature.

Sensor measures temperature → Feedback → Thermostat compares and adjusts cooling.

2. Car Cruise Control:

Input: Desired speed.

Output: Actual car speed.

Speed sensor provides feedback → Controller adjusts throttle to maintain constant speed even on slopes.

Advantages

- **High accuracy:** Output remains close to desired input.
- **Automatic error correction:** Corrects disturbances (e.g., external temperature changes, load changes).
- **Adaptability:** Adjusts automatically to changes in system dynamics or environment.

Disadvantages

- **More complex:** Requires sensors, feedback mechanisms, and error comparator.
- **Expensive:** Higher cost due to additional components and design complexity.
- **Stability issues:** If not properly designed, may oscillate or become unstable.

Open-loop vs Closed-loop

Feature	Open-loop System	Closed-loop System
Definition	Control action independent of output	Control action depends on output feedback
Block Diagram	Input → Controller → Plant → Output	Input → Comparator → Controller → Plant → Output → Feedback
Example	Electric iron, washing machine timer	AC thermostat, cruise control
Accuracy	Low	High
Error Correction	Not possible	Possible
Cost & Complexity	Simple and cheap	Complex and costly
Stability	Always stable	May face stability issues

Importance of Feedback in Control

- Reduces error between actual output and desired output.
- Improves system stability and accuracy.
- Provides adaptability to disturbances and changes in environment.
- Essential in automation, robotics, process control, and modern smart systems.

Actuation Systems

Actuation systems are the elements of control systems which are responsible for transforming the output of a microcontrollers or microprocessor or control system into a controlling action on machine or device. e.g., mechanical, pneumatic, hydraulic, and electrical actuation systems.

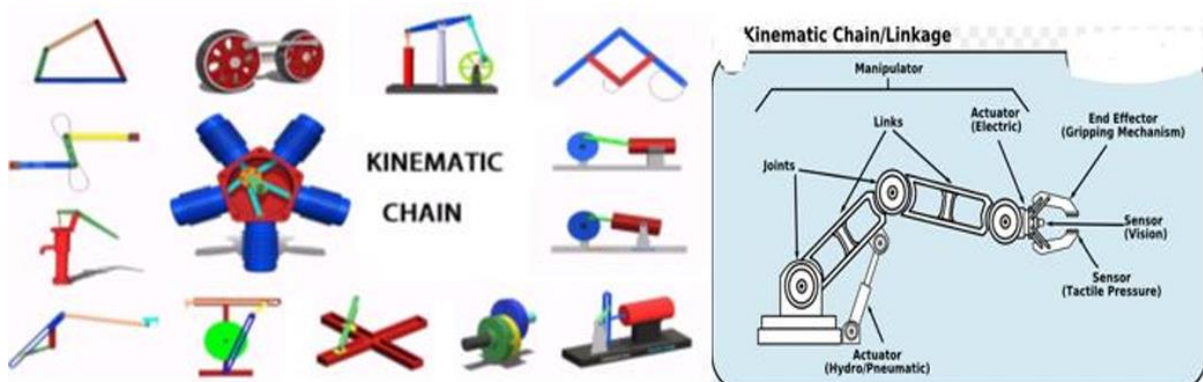
Mechanical Actuation Systems

Mechanical actuation systems are mechanisms that **convert input motion/force into desired output motion/force**. They are essential in machines, automation, robotics, and mechanical devices.

Some mechanical components used for actuation purpose are given here:

1. Kinematic Chains

A **kinematic chain** is a combination of rigid bodies (links) connected by joints in such a way that there is a definite relative motion between them. Each link can have translational or rotational movement depending on joints.



Types of Kinematic Chains

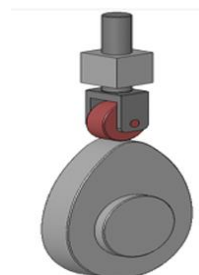
1. **Closed Chain** – Forms a closed loop (e.g., four-bar chain).
2. **Open Chain** – Does not form a loop (e.g., robotic arm).

Applications: Linkages in engines, robots, cranes, and mechanical arms.

2. Cam Mechanisms

A **cam mechanism** consists of a **cam** (driving member) and a **follower** (driven member). The cam converts **rotary motion into reciprocating or oscillating motion**.

A cam is a rotating or sliding piece in a mechanical linkage used especially in transforming rotary motion into linear motion. It is often a part of a rotating wheel (e.g. an eccentric wheel) or shaft (e.g. a cylinder with an



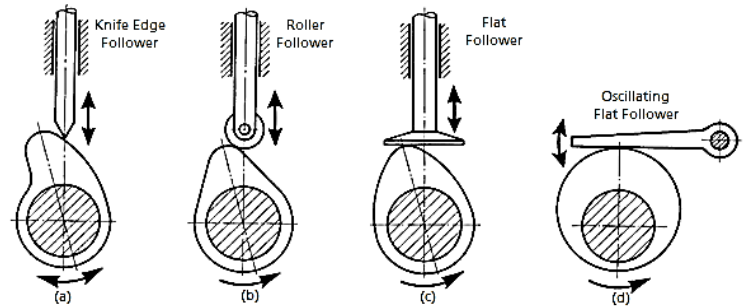
irregular shape) that strikes a lever at one or more points on its circular path. The cam can be a simple tooth, as is used to deliver pulses of power to a steam hammer, for example, or an eccentric disc or other shape that produces a smooth reciprocating (back and forth) motion in the follower, which is a lever making contact with the cam.

Types of Followers

- Knife-edge follower
- Roller follower
- Flat-faced follower

Applications

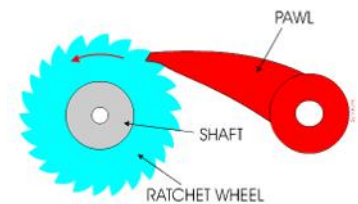
- Valve operation in IC engines.
- Textile machinery.
- Automation equipment.



3. Ratchet Mechanisms

A **ratchet mechanism** allows **motion in one direction only** while preventing motion in the reverse direction. It consists of a ratchet wheel and a pawl.

A ratchet is a fairly simple mechanism that only allows a gear to turn in one direction. • A ratchet system consists of a gear (sometimes the teeth are different than the standard profile) with a small lever or latch that rotates about a pivot point and catches in the teeth of the gear. • The latch is designed and oriented such that if the gear were to turn in one direction, the gear could spin freely and the latch would be pushed up by the teeth, but if the gear were to spin in the other direction, the latch would catch in the teeth of the gear and prevent it from moving. • Ratchets are useful in a variety of applications, because they allow force to be applied in one direction but not the other. • Bicycles, some wrenches, screw gauge, etc.



Applications: Clock winding mechanism, Lifting jacks and hoists, Bicycle freewheel system.

4. Gears

Gears are toothed wheels that transmit power and motion between shafts by meshing without slip. Gears are used where precise motion and high torque transmission are required.

The teeth of a gear on one axis mesh with the teeth of a gear on another, thus creating a relationship between the rotation of the two axes. When one axis is spun, the other will too. Two gears of different sizes will make their two axes spin at different speeds.

Types of Gears



1. **Spur Gear** – Teeth parallel to axis, used for parallel shafts.
2. **Helical Gear** – Teeth inclined, smoother and quieter than spur.
3. **Bevel Gear** – Conical shape, for intersecting shafts.
4. **Worm Gear** – Screw-like worm meshes with a worm wheel, for high reduction ratios.
5. **Rack and Pinion** – Converts rotary motion into linear motion.
6. **Planetary Gear (Epicyclic)** – Compact, high torque transmission, used in gearboxes.

Applications: Automobiles, gearboxes, watches, turbines, industrial machines.

5. Belts

Belts are flexible elements that transmit power between two rotating shafts using pulleys. A belt is a loop of flexible material used to link two or more rotating shafts mechanically, most often parallel. Belts may be used as a source of motion, to transmit power efficiently or to track relative movement. Belts are looped over pulleys and may have a twist between the pulleys, and the shafts may or may not be parallel.

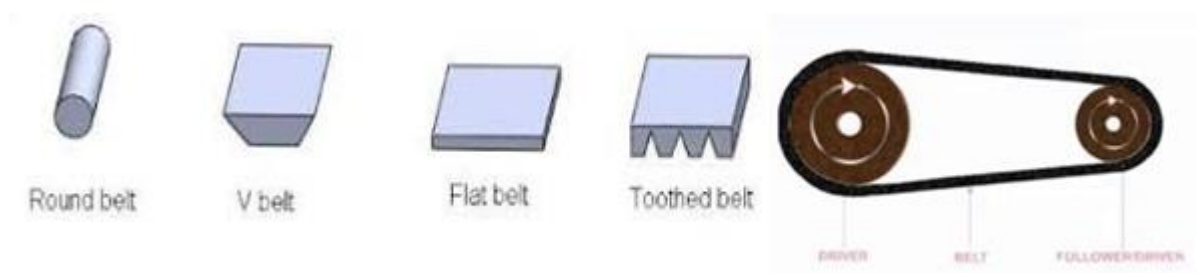
Types of Belts

Flat Belt – Transmits moderate power, used in conveyors.

V-Belt – Wedge action increases friction, commonly used in automobiles.

Timing Belt (Toothed Belt) – Prevents slip, used for synchronous power transmission.

Round Belt – Flexible, used in small appliances.



6. Bearings

Bearings support and guide rotating shafts, reducing friction and wear. A bearing is a machine element that constrains relative motion to only the desired motion, and reduces friction between moving parts. Bearings support the rotating shafts of the wheels, gears, turbines, rotors, etc. in those

machines, allowing them to rotate more smoothly. They fulfil the following two major functions. Function 1: Reduce friction and make rotation more smooth. Function 2: Protect the part that supports the rotation, and maintain the correct position for the rotating shaft.

Types of Bearings

Sliding Contact Bearings – Surface-to-surface contact (e.g., bush bearing).

Rolling Contact Bearings – Balls or rollers reduce friction.

Ball Bearings – Handle both radial and thrust loads.

Roller Bearings – Cylindrical/needle rollers for higher load capacity.

Tapered Roller Bearings – Used in vehicle wheel hubs.

Applications: Automobiles (wheel hubs, engines), turbines, pumps, electric motors, industrial machines.

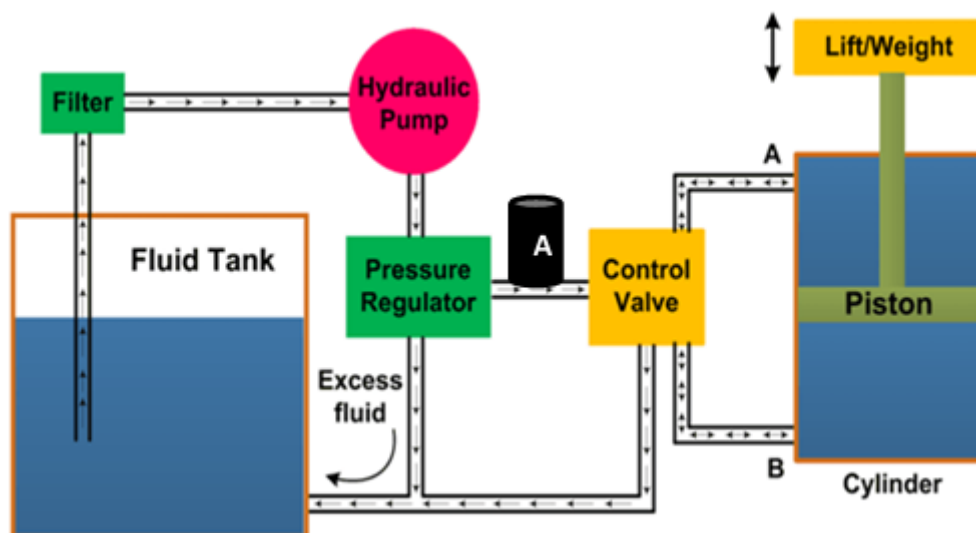


Overview of Fluid Power Systems: Fluid power systems use **pressurized fluids** (liquids or gases) to generate, control, and transmit power.

- Hydraulic systems → use incompressible fluids (oil)
- Pneumatic systems → use compressible fluids (air)

Hydraulic systems

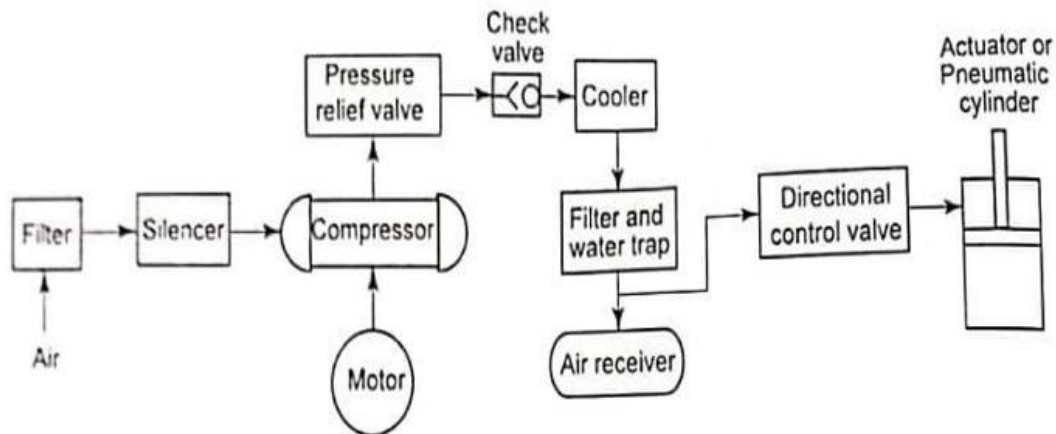
It is a kind of enclosed fluid-based system using pressurized incompressible liquids as transmission media and is called as hydraulic system. The hydraulic system works on the principle of Pascal's law. Enclosed fluid systems can provide both linear as well as rotary motions. The hydraulic systems consists of storage tank, filter, hydraulic pump, non-return valve, accumulator, pressure regulator, direction control valves, hydraulic cylinder, piston, and leak-proof fluid flow pipelines. The storage/fluid tank is a reservoir for the liquid used as a transmission media. The liquid used is generally high-density incompressible oil. It is filtered to remove dust or any other unwanted particles and then pumped by the hydraulic pump operated by a motor. The capacity of the pump depends on the hydraulic system design.



A non-return valve is used to make sure no amount of liquid returns back to the pumping system. The pumps generally deliver constant volume in each revolution of the pump shaft. Therefore, the fluid pressure can increase indefinitely at the dead-end of the piston until the system fails. The pressure regulator / relief valve is used to avoid such circumstances which redirect the excess fluid back to the storage tank. In case the pump delivers non-uniform fluid supply, an accumulator needs to be used to regulate and control the amount of fluid flow. It helps in minimizing the pressure fluctuations in supply line. The movement of the piston is controlled by changing the liquid flow from port A and port B. The cylinder movement is controlled by using a control valve which directs the fluid flow. The fluid pressure line is connected to the port B to raise the piston and it is connected to port A to lower down the piston. The valve can also stop the fluid flow in any of the ports. The leak-proof piping is also important due to safety, environmental hazards, and economic aspects.

Pneumatic systems

In pneumatic systems, force is produced by gas. It is mainly by air pressure acting on the surface of a piston or valve. Compressed air is produced in a compressor and stored in a receiver. From compressor, it is sent to valves which control the direction of fluid flow. Also, flow control valves control the amount of power by the cylinder.



The fresh atmosphere air is not sent directly to the compressor to use in pneumatic systems. First, it is filtered atmosphere air is supplied to the compressor through silencer to reduce noise level. Then it is compressed. Pressure relief valve is used to avoid the damage of compressor due to excess pressure raise in the system. Check valve is a one-way valve that allows pressurized air to enter the pneumatic system, but prevents backflow and loss of pressure into the compressor when it is stopped. The cooler is used to cool the compressed air which is usually very hot. The filter is used to remove contamination in the compressed air and water trap is used to remove water particles. The pressurized air is stored in a device called an air receiver, preventing surges in pressure and relieving the duty cycle of the compressor. Directional control valves are used to control flow of pressurized air from the source to the selected port. These valves can be actuated either manually or electrically. Actuator or pneumatic cylinder converts energy stored in the compressed air into mechanical motion.

Major Components

Common Components (Both Systems)

1. **Actuator** – Converts pressure energy into mechanical motion (linear or rotary).
2. **Pump/Compressor** – Generates fluid flow.
3. **Reservoir/Air Tank** – Stores fluid or air.
4. **Valves** – Control pressure, direction, and flow rate.
5. **Piping and Fittings** – Transmit the working fluid.
6. **Filter** – Removes impurities.
7. **Pressure Gauge** – Monitors system pressure.

Hydraulic System Components

1. **Hydraulic Pump** – Converts mechanical energy into fluid power.
Types: Gear pump, vane pump, piston pump.
2. **Hydraulic Actuator** – Converts fluid power into motion.
Types:
 - **Cylinders** (linear motion)
 - **Hydraulic motors** (rotary motion)
3. **Control Valves:**
 - **Direction control valve (DCV)** – decides actuator movement direction.
 - **Pressure relief valve** – prevents overpressure.
 - **Flow control valve** – controls speed.
4. **Reservoir (Tank):** Stores oil and dissipates heat.
5. **Hydraulic Fluid:** Acts as a medium for power transmission and lubrication.

Pneumatic System Components

1. **Air Compressor:** Compresses atmospheric air.
Types: Reciprocating, rotary screw, centrifugal.
2. **Air Preparation Unit (FRL Unit):**
 - **Filter** – removes dust and moisture.
 - **Regulator** – controls pressure.
 - **Lubricator** – adds oil mist for lubrication.
3. **Pneumatic Actuators:**
 - **Cylinders (linear)** – single acting or double acting.
 - **Air motors (rotary)** – vane or piston type.
4. **Valves:**
 - **Directional control valve (DCV)** – controls airflow direction.
 - **Flow control valve** – adjusts speed.
 - **Quick exhaust valve** – increases piston return speed.
5. **Air Lines and Fittings:** Connect various components.

Advantages and Disadvantages

Hydraulic System

Advantages:

- High force and torque output.
- Smooth and precise control.
- Can handle heavy loads.
- Compact design for large power transmission.

Disadvantages:

- Oil leakage causes contamination.
- Complex maintenance.

- Sensitive to temperature variations.
- Slower response compared to pneumatics.

Pneumatic System

Advantages:

- Clean and environment-friendly.
- Simple and fast operation.
- Safe (no fire hazard).
- Easy to maintain and inexpensive.

Disadvantages:

- Limited force due to low pressure.
- Compressibility of air reduces precision.
- Noise from exhaust air.
- Requires drying and filtration.

Applications

Hydraulic Systems	Pneumatic Systems
Excavators, cranes, loaders	Packaging machines
Hydraulic presses and lifts	Pneumatic tools (drills, grinders)
Aircraft control systems	Automation and robotics
Injection molding machines	Pick-and-place units
Power steering, brakes	Air brakes (trains, buses)

Comparison between Hydraulic and Pneumatic Systems

Feature	Hydraulic System	Pneumatic System
Working medium	Oil (incompressible)	Air (compressible)
Operating pressure	50–500 bar	5–10 bar
Power output	Very high	Moderate
Speed of operation	Slower	Faster
Accuracy and control	High precision	Moderate precision
Cleanliness	Can leak and cause mess	Clean and safe
Cost	Higher initial cost	Lower cost
Maintenance	Requires oil maintenance	Easier maintenance
Applications	Heavy-duty (presses, excavators)	Light-duty (automation, pick-and-place)

Various types of Valves

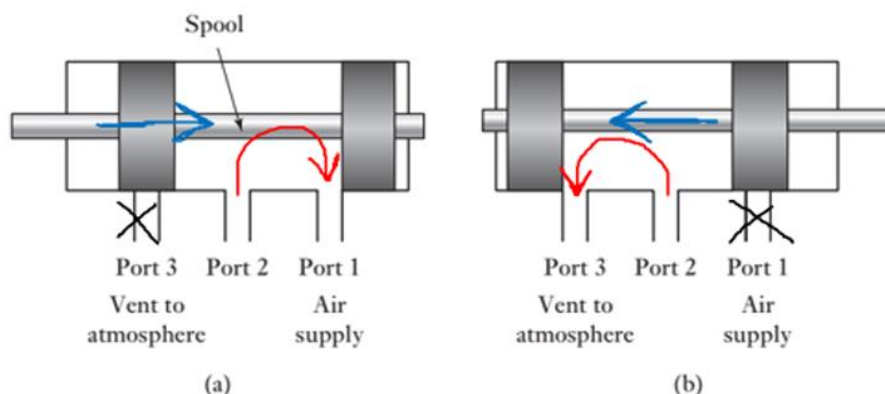
- Valves are used with hydraulic and pneumatic systems to direct and regulate the fluid flow.
- There are basically just two forms of valve, the finite position and the infinite position valves.
- The finite position valves are ones where the action is just to allow or block fluid flow and so can be used to switch actuators on or off. They can be used for directional control to switch the flow from one path to another and so from one actuator to another.
- The infinite position valves are able to control flow anywhere between fully on and fully off and so are used to control varying actuator forces or the rate of fluid flow for a process control situation.

Directional Control Valves

They are not intended to vary the rate of flow of fluid but are either completely open or completely closed, i.e. on/off devices. They might be activated to switch the fluid flow direction by means of mechanical, electrical or fluid pressure signals. Some common type of directional control valve is the SPOOL VALVE, Poppet valve.

Spool Valve

A spool moves horizontally within the valve body to control the flow. In (a) the air supply is connected to port 1 and port 3 is closed. Thus the device connected to port 2 can be pressurised. When the spool is moved to the left the air supply is cut off and port 2 is connected to port 3. Port 3 is a vent to the atmosphere and so the air pressure in the system attached to port 2 is vented. Thus the movement of the spool has allowed the air firstly to flow into the system and then be reversed and flow out of the system.



Poppet Valve

Common form of pressure relief valve is the poppet valve. If pressure at the inlet of relief valve is not enough to overcome the spring force, in that situation poppet will remain seated over its seat and will not allow the flow of fluid through it and therefore relief valve will be closed and will not allow flow through it.

When pressure at the inlet of relief valve increases and overcomes the spring force which was adjusted with the help of pressure adjusting screw displayed at the top of relief valve, in that situation poppet will leave the seat and will permit the flow of fluid through relief valve.

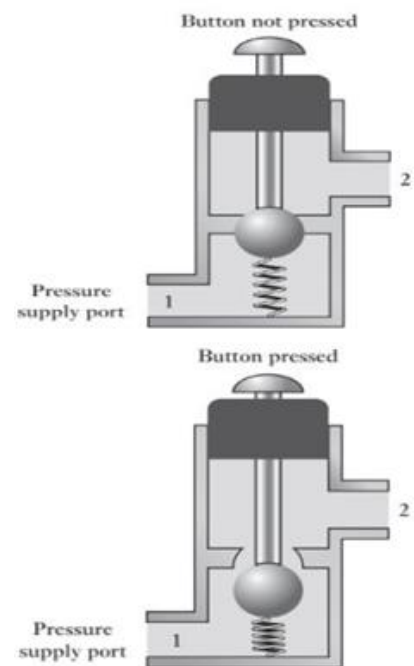
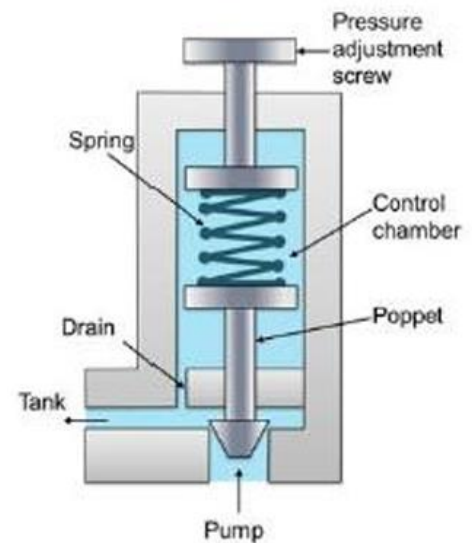
Hence, if pressure at inlet of relief valve overcomes the setting pressure of relief valve, then fluid will flow to hydraulic reservoir through relief valve and therefore further increment in pressure in pressure line will be avoided.

It may be operated manually also, using a push-button on its top. This valve is normally in the closed condition, there being no connection between port 1 to which the pressure supply is connected and port 2 to which the system is connected.

In poppet valves, balls, discs or cones are used in conjunction with valve seats to control the flow.

In the figure a ball is shown. When the push-button is depressed, the ball is pushed out of its seat and flow occurs as a result of port 1 being connected to port 2.

When the button is released, the spring forces the ball back up against its seat and so closes off the flow.

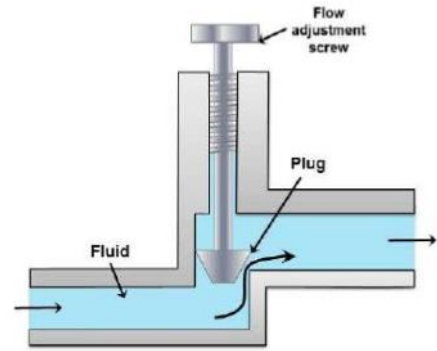


Flow Control Valves:

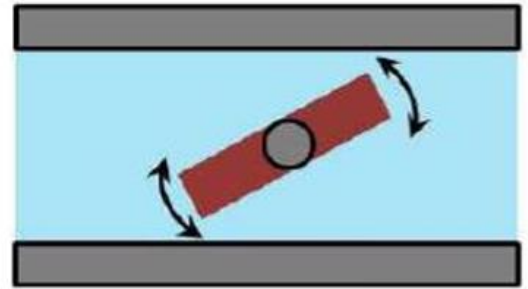
Flow control valves in hydraulics are used for controlling the volume of fluid which is supplied to different parts of the hydraulic system. The speed of the actuators which is being used in the hydraulic circuits can be controlled by regulating the fluid flow, to carry out this regulation operation we need devices and the device which is providing that regulation operation is flow regulation valves.

Types of flow control valves

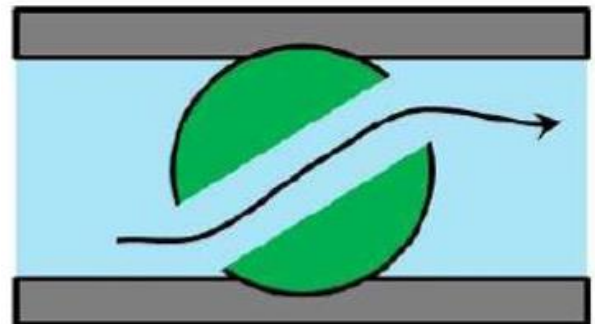
Plug or glove valve : The plug valve is quite commonly used valve. It is also termed as glove valve. This valve has a plug which can be adjusted in vertical direction by setting flow adjustment screw. The adjustment of plug alters the orifice size between plug and valve seat.



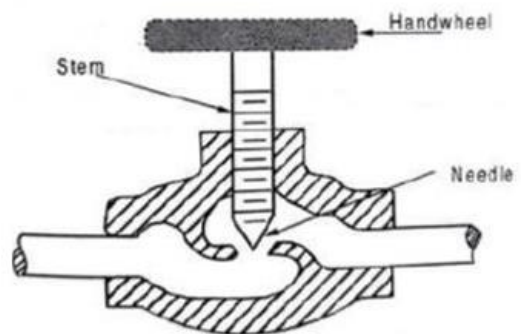
Butterfly valve : It consists of a disc which can rotate inside the pipe. The angle of disc determines the restriction. Butterfly valve can be made to any size and is widely used to control the flow of gas. The resilient butterfly valve uses the flexibility of rubber and has the lowest pressure rating.



Ball valve : This type of flow control valve uses a ball rotated inside a machined seat. The ball has a through hole as shown in figure. It has very less leakage in its shut-off condition. These valves are durable and usually work perfectly for many years. They are excellent choice for shutoff applications. They do not offer fine control which may be necessary in throttling applications. These valves are widely used in industries because of their versatility, high supporting pressures (up to 1000 bar) and temperatures (up to 250°C). They are easy to repair and operate.



Pin or needle valve : Needles valves provide finer control of flow in small diameter pipes. They have sharp pointed conical disc and matching seat. Needle valves are normally made up of steel bar. These are also used as stop valves in hydraulic circuit to shut off the flow of fluid from one part of a circuit to another part. Needle valves are suitable for throttling i.e., the flow area is slowly reduced as the valve is closed, gradually reducing the quantity of fluid passing through the valve.



Rotary Actuators: Rotary actuators convert fluid pressure into rotational motion.

Types:

1. Hydraulic Motor:

- Provides continuous rotary motion.
- Types: Gear, Vane, Piston motors.
- Used in winches, conveyors, rotary tables.

2. Pneumatic Rotary Actuator:

- Provides limited angular movement (e.g., 90°, 180°).
- Types: Vane type, Rack-and-pinion type.
- Used for valve operation, pick-and-place systems.

Pressure sequence valves:

- These valves are used to sense the pressure of an external line and give a signal when it reaches some preset value. With the pressure-limiting valve, the limiting pressure is set by the pressure at the inlet to the valve.
- We can adapt such a valve to give a sequence valve.
- This can be used to allow flow to occur to some part of the system when the pressure has risen to the required level.
- For example, in an automatic machine we might require some operation to start when the clamping pressure applied to a workpiece is at some particular value.

